Human review packet: PKG25NoFreeLunch

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: PKG25NoFreeLunch
Paper id: PKG25NoFreeLunch
Source version: AAAI 2025, cross-checked against arXiv 2411.15230 source archive SHA256 da7868015db272f3c61cdc88c4e8492b8888f8363332215aa59c509471831c56
Source record: audit/paper_statement_map.json
Rows: 5
Generated: 2026-09-06
Source URL: official source

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper PKG25NoFreeLunch --serve
```

Open http://127.0.0.1:8765/ in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

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Paper-specific semantic prerequisites

PKG25NoFreeLunch.JointLawCollaborationSetting

Paper-local declaration:
PKG25NoFreeLunch.JointLawCollaborationSetting
Source locator: cited publication, lines 13-19

Verbatim paper-source connection

```
We may now define collaboration settings.
\begin{definition}A \textbf{collaboration setting} is an ordered tuple
\begin{equation}
    S=(\DD, P_1, \cdots, P_n),
\end{equation}
where $\DD$ is a probability distribution over $\XX\times \{0,1\}$ and $P_1,\cdots,P_n$ are each calibrated predictors on $\DD$.
\end{definition}

[Next verbatim source excerpt]

A \vocab{predictor} is a function $P: \XX\rightarrow [0,1]$. A predictor $P$ is \vocab{calibrated} on $\DD$ if
\begin{equation}
    \Pr_{(X,Y)\sim \DD}[Y=1\,|\,P(X)=p] = p
\end{equation}
for all $p\in [0,1]$ (more precisely, for all $p\in \text{Image}(P)$).
```

Lean-elaborated paper signature

```
type:
ℕ → Type 1

constructor PKG25NoFreeLunch.JointLawCollaborationSetting.mk:
{n : ℕ} →
  (X : Type) →
    [measurableSpaceX : MeasurableSpace X] →
      (joint : MeasureTheory.Measure (X × PKG25NoFreeLunch.Label)) →
        MeasureTheory.IsProbabilityMeasure joint →
          (pred : Fin n → X → ℝ) →
            (∀ (i : Fin n) (x : X), 0 ≤ pred i x ∧ pred i x ≤ 1) →
              (∀ (i : Fin n), Measurable (pred i)) →
                (∀ (i : Fin n) (A : Set ℝ),
                    MeasurableSet A →
                      ∫ (z : X × PKG25NoFreeLunch.Label),
                          {z : X × PKG25NoFreeLunch.Label | pred i z.1 ∈ A ∧ z.2 = true}.indicator
                            (fun (x : X × PKG25NoFreeLunch.Label) => 1) z ∂joint =
                        ∫ (z : X × PKG25NoFreeLunch.Label),
                          {z : X × PKG25NoFreeLunch.Label | pred i z.1 ∈ A}.indicator
                            (fun (z : X × PKG25NoFreeLunch.Label) => pred i z.1) z ∂joint) →
                  PKG25NoFreeLunch.JointLawCollaborationSetting n

field X:
{n : ℕ} → PKG25NoFreeLunch.JointLawCollaborationSetting n → Type

field measurableSpaceX:
{n : ℕ} → (self : PKG25NoFreeLunch.JointLawCollaborationSetting n) → MeasurableSpace self.X

field joint:
{n : ℕ} →
  (self : PKG25NoFreeLunch.JointLawCollaborationSetting n) → MeasureTheory.Measure (self.X × PKG25NoFreeLunch.Label)

field isProbability:
∀ {n : ℕ} (self : PKG25NoFreeLunch.JointLawCollaborationSetting n), MeasureTheory.IsProbabilityMeasure self.joint

field pred:
{n : ℕ} → (self : PKG25NoFreeLunch.JointLawCollaborationSetting n) → Fin n → self.X → ℝ

field pred_range:
∀ {n : ℕ} (self : PKG25NoFreeLunch.JointLawCollaborationSetting n) (i : Fin n) (x : self.X),
  0 ≤ self.pred i x ∧ self.pred i x ≤ 1

field pred_measurable:
∀ {n : ℕ} (self : PKG25NoFreeLunch.JointLawCollaborationSetting n) (i : Fin n), Measurable (self.pred i)

field calibrated_events:
∀ {n : ℕ} (self : PKG25NoFreeLunch.JointLawCollaborationSetting n) (i : Fin n) (A : Set ℝ),
  MeasurableSet A →
    ∫ (z : self.X × PKG25NoFreeLunch.Label),
        {z : self.X × PKG25NoFreeLunch.Label | self.pred i z.1 ∈ A ∧ z.2 = true}.indicator
          (fun (x : self.X × PKG25NoFreeLunch.Label) => 1) z ∂self.joint =
      ∫ (z : self.X × PKG25NoFreeLunch.Label),
        {z : self.X × PKG25NoFreeLunch.Label | self.pred i z.1 ∈ A}.indicator
          (fun (z : self.X × PKG25NoFreeLunch.Label) => self.pred i z.1) z ∂self.joint
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The structure contains exactly an input space, a probability law on input-label pairs, and n measurable [0,1]-valued calibrated predictors. Its event identity is the measure-theoretic version of the source's conditional calibration formula, agreeing with that formula on positive-mass prediction fibers and avoiding an arbitrary conditional-probability choice on null fibers.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.JointLawCollaborationSetting.agentAccuracy_eq_predictionCertainty

Paper-local declaration:
PKG25NoFreeLunch.JointLawCollaborationSetting.agentAccuracy_eq_predictionCertainty
Source locator: cited publication, lines 21-24

Verbatim paper-source connection

```
The predictor $P_i$ induces the classifier $\Yhat_i: x\mapsto \round{P_i(x)},$ where $\round{\cdot}$ is the rounding operator (without loss of generality,
↪    set $\round{0.5}=1$). $\Yhat_i$ achieves 0-1 accuracy
\begin{equation}
    \acc_i(S) := \EE_{(X,Y)\sim \DD}[\max\{P_i(X), 1 - P_i(X)\}].
\end{equation}
```

Lean-elaborated paper signature

```
∀ {n : ℕ} (S : PKG25NoFreeLunch.JointLawCollaborationSetting n) (i : Fin n),
  S.agentAccuracy i = ∫ (z : S.X × PKG25NoFreeLunch.Label), max (S.pred i z.1) (1 - S.pred i z.1) ∂S.joint
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The theorem equates the 0-1 accuracy of the rounded predictor with the source's displayed expectation of max(P_i(X), 1-P_i(X)). It derives the identity from the separately reviewed calibration semantics and uses the source's tie-up rounding convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.JointLawCollaborationSetting.agentClassifier

Paper-local declaration:
PKG25NoFreeLunch.JointLawCollaborationSetting.agentClassifier
Source locator: cited publication, lines 21-24

Verbatim paper-source connection

```
The predictor $P_i$ induces the classifier $\Yhat_i: x\mapsto \round{P_i(x)},$ where $\round{\cdot}$ is the rounding operator (without loss of generality,
↪    set $\round{0.5}=1$). $\Yhat_i$ achieves 0-1 accuracy
\begin{equation}
    \acc_i(S) := \EE_{(X,Y)\sim \DD}[\max\{P_i(X), 1 - P_i(X)\}].
\end{equation}
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → (S : PKG25NoFreeLunch.JointLawCollaborationSetting n) → Fin n → S.X → PKG25NoFreeLunch.Label

value:
fun {n : ℕ} (S : PKG25NoFreeLunch.JointLawCollaborationSetting n) (i : Fin n) (x : S.X) =>
  PKG25NoFreeLunch.roundProb (S.pred i x)
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The declaration maps an input x to round(P_i(x)), exactly the source-induced individual classifier. The referenced roundProb declaration separately fixes the source convention round(0.5)=1.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.JointLawCollaborationSetting.sourcePredictionProfile

Paper-local declaration:
PKG25NoFreeLunch.JointLawCollaborationSetting.sourcePredictionProfile
Source locator: cited publication, lines 31-35

Verbatim paper-source connection

```
Here, $\CC$ should be interpreted as a function that takes in $n$ predicted probabilities and returns a 0-1 classification. Given a collaboration setting
↪    $S=(\DD, P_1, \cdots, P_n)$, a collaboration strategy $\CC$ induces the classifier
\begin{equation}
    \Yhat_\CC: \XX \rightarrow \{0,1\},\quad x \mapsto \CC(P_1(x),\cdots,P_n(x)).
\end{equation}
Let $\acc_\CC(S)$ denote the 0-1 accuracy of $\Yhat_\CC$ on $\DD$. In other words, for each $x\in \XX$, each agent $i$ produces a predicted probability
↪    $P_i(x).$ The collaboration strategy aggregates these predictions into a binary classification.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → (S : PKG25NoFreeLunch.JointLawCollaborationSetting n) → S.X → PKG25NoFreeLunch.UnitCubeProfile n

value:
fun {n : ℕ} (S : PKG25NoFreeLunch.JointLawCollaborationSetting n) (x : S.X) =>
  ⟨fun (i : Fin n) => S.pred i x, fun (i : Fin n) => S.pred_range i x⟩
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The target packages exactly the n displayed predictor values with their coordinatewise [0,1] bounds, which is the prediction profile supplied to the induced classifier.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.JointLawCollaborationSetting.sourceStrategyAccuracy

Paper-local declaration:
PKG25NoFreeLunch.JointLawCollaborationSetting.sourceStrategyAccuracy
Source locator: cited publication, lines 35

Verbatim paper-source connection

```
Let $\acc_\CC(S)$ denote the 0-1 accuracy of $\Yhat_\CC$ on $\DD$. In other words, for each $x\in \XX$, each agent $i$ produces a predicted probability
↪    $P_i(x).$ The collaboration strategy aggregates these predictions into a binary classification.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → PKG25NoFreeLunch.JointLawCollaborationSetting n → PKG25NoFreeLunch.SourceCollaborationStrategy n → ℝ

value:
fun {n : ℕ} (S : PKG25NoFreeLunch.JointLawCollaborationSetting n)
    (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) =>
  S.classifierAccuracy (S.sourceStrategyClassifier C)
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The target is the 0-1 classifier accuracy of the displayed induced source-domain strategy classifier. Its expectation-definedness use is governed by the displayed approved integrability convention.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.JointLawCollaborationSetting.sourceStrategyClassifier

Paper-local declaration:
PKG25NoFreeLunch.JointLawCollaborationSetting.sourceStrategyClassifier
Source locator: cited publication, lines 31-35

Verbatim paper-source connection

```
Here, $\CC$ should be interpreted as a function that takes in $n$ predicted probabilities and returns a 0-1 classification. Given a collaboration setting
↪    $S=(\DD, P_1, \cdots, P_n)$, a collaboration strategy $\CC$ induces the classifier
\begin{equation}
    \Yhat_\CC: \XX \rightarrow \{0,1\},\quad x \mapsto \CC(P_1(x),\cdots,P_n(x)).
\end{equation}
Let $\acc_\CC(S)$ denote the 0-1 accuracy of $\Yhat_\CC$ on $\DD$. In other words, for each $x\in \XX$, each agent $i$ produces a predicted probability
↪    $P_i(x).$ The collaboration strategy aggregates these predictions into a binary classification.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} →
  (S : PKG25NoFreeLunch.JointLawCollaborationSetting n) →
    PKG25NoFreeLunch.SourceCollaborationStrategy n → S.X → PKG25NoFreeLunch.Label

value:
fun {n : ℕ} (S : PKG25NoFreeLunch.JointLawCollaborationSetting n) (C : PKG25NoFreeLunch.SourceCollaborationStrategy n)
    (x : S.X) =>
  C (S.sourcePredictionProfile x)
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: For every input x, the target applies C to the exact source-domain profile of P_1(x), ..., P_n(x), yielding the displayed induced binary classifier.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.SourceCollaborationStrategy

Paper-local declaration:

PKG25NoFreeLunch.SourceCollaborationStrategy
Source locator: cited publication, lines 27-35

Verbatim paper-source connection

```
A collaboration strategy is a way to combine the predicted probabilities of each agent to make a classification.
\begin{definition}
A \textbf{collaboration strategy} is a deterministic function $\CC: [0,1]^n \rightarrow \{0,1\}.$
\end{definition}
Here, $\CC$ should be interpreted as a function that takes in $n$ predicted probabilities and returns a 0-1 classification. Given a collaboration setting
↪    $S=(\DD, P_1, \cdots, P_n)$, a collaboration strategy $\CC$ induces the classifier
\begin{equation}
    \Yhat_\CC: \XX \rightarrow \{0,1\},\quad x \mapsto \CC(P_1(x),\cdots,P_n(x)).
\end{equation}
Let $\acc_\CC(S)$ denote the 0-1 accuracy of $\Yhat_\CC$ on $\DD$. In other words, for each $x\in \XX$, each agent $i$ produces a predicted probability
↪    $P_i(x).$ The collaboration strategy aggregates these predictions into a binary classification.
```

Lean-elaborated paper signature

```
type:
ℕ → Type

value:
fun (n : ℕ) => PKG25NoFreeLunch.UnitCubeProfile n → PKG25NoFreeLunch.Label
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: UnitCubeProfile is the displayed coordinatewise closed unit cube and Label is Bool, so the target is exactly a deterministic function [0,1]^n to {0,1}.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.SourceConstantOnHalfSlice

Paper-local declaration:
PKG25NoFreeLunch.SourceConstantOnHalfSlice
Source locator: cited publication, lines 42-54

Verbatim paper-source connection

```
\begin{definition}\label{def:non-collaborative}
    A collaboration strategy $\CC:[0,1]^n\rightarrow \{0,1\}$ is \textbf{non-collaborative} if there exists $k\in [n]$ and $\alpha\in \{0,1\}$ such that
    \begin{equation}
    \CC(p_1,p_2,\cdots,p_n) =
        \begin{cases}
        \round{p_k} &\quad \text{$p_k\neq \frac{1}{2}$} \\
            \alpha &\quad \text{$p_k=\frac{1}{2}$}
        \end{cases}
    \end{equation}
    for all $(p_1,p_2,\cdots,p_n)\in (0,1)^n$.
\end{definition}

A collaboration strategy is non-collaborative if it essentially always defers to the classification of a single agent $k\in [n]$. The definition admits two
↪    exceptions. First, when $p_k = \frac{1}{2}$, the collaboration may select either $0$ or $1$, but must always select the same such value; this choice
↪    has no effect on the accuracy of the classifier, so this exception can be essentially ignored. Second, the collaboration strategy need not select
↪    $\round{p_k}$ when $(p_1,p_2,\cdots,p_n)\notin (0,1)^n$---i.e., when $p_i\in \{0,1\}$ for some $i\in [n]$; this exception is somewhat more
↪    interesting, and we will return to it after stating our main result.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → PKG25NoFreeLunch.SourceCollaborationStrategy n → Fin n → PKG25NoFreeLunch.Label → Prop

value:
fun {n : ℕ} (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) (k : Fin n) (alpha : PKG25NoFreeLunch.Label) =>
  ∀ (p : PKG25NoFreeLunch.UnitCubeProfile n), PKG25NoFreeLunch.Interior ↑p → ↑p k = 1 / 2 → C p = alpha
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The target quantifies exactly over interior cube profiles with p_k = 1/2 and requires one fixed alpha, precisely the source half-slice clause.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.SourceDefersAwayFromHalf

Paper-local declaration:
PKG25NoFreeLunch.SourceDefersAwayFromHalf
Source locator: cited publication, lines 42-54

Verbatim paper-source connection

```
\begin{definition}\label{def:non-collaborative}
    A collaboration strategy $\CC:[0,1]^n\rightarrow \{0,1\}$ is \textbf{non-collaborative} if there exists $k\in [n]$ and $\alpha\in \{0,1\}$ such that
    \begin{equation}
    \CC(p_1,p_2,\cdots,p_n) =
        \begin{cases}
        \round{p_k} &\quad \text{$p_k\neq \frac{1}{2}$} \\
            \alpha &\quad \text{$p_k=\frac{1}{2}$}
        \end{cases}
    \end{equation}
    for all $(p_1,p_2,\cdots,p_n)\in (0,1)^n$.
\end{definition}

A collaboration strategy is non-collaborative if it essentially always defers to the classification of a single agent $k\in [n]$. The definition admits two
↪    exceptions. First, when $p_k = \frac{1}{2}$, the collaboration may select either $0$ or $1$, but must always select the same such value; this choice
↪    has no effect on the accuracy of the classifier, so this exception can be essentially ignored. Second, the collaboration strategy need not select
↪    $\round{p_k}$ when $(p_1,p_2,\cdots,p_n)\notin (0,1)^n$---i.e., when $p_i\in \{0,1\}$ for some $i\in [n]$; this exception is somewhat more
↪    interesting, and we will return to it after stating our main result.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → PKG25NoFreeLunch.SourceCollaborationStrategy n → Fin n → Prop

value:
fun {n : ℕ} (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) (k : Fin n) =>
  ∀ (p : PKG25NoFreeLunch.UnitCubeProfile n),
    PKG25NoFreeLunch.Interior ↑p → ↑p k ≠ 1 / 2 → C p = PKG25NoFreeLunch.roundProb (↑p k)
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The target quantifies exactly over interior cube profiles with p_k != 1/2 and requires C(p) = round(p_k), precisely the source off-half clause.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.SourceNonCollaborative

Paper-local declaration:
PKG25NoFreeLunch.SourceNonCollaborative
Source locator: cited publication, lines 42-54

Verbatim paper-source connection

```
\begin{definition}\label{def:non-collaborative}
  A collaboration strategy $\CC:[0,1]^n\rightarrow \{0,1\}$ is \textbf{non-collaborative} if there exists $k\in [n]$ and $\alpha\in \{0,1\}$ such that
  \begin{equation}
  \CC(p_1,p_2,\cdots,p_n) =
    \begin{cases}
    \round{p_k} &\quad \text{$p_k\neq \frac{1}{2}$} \\
      \alpha &\quad \text{$p_k=\frac{1}{2}$}
    \end{cases}
  \end{equation}
  for all $(p_1,p_2,\cdots,p_n)\in (0,1)^n$.
\end{definition}

A collaboration strategy is non-collaborative if it essentially always defers to the classification of a single agent $k\in [n]$. The definition admits two
↪   exceptions. First, when $p_k = \frac{1}{2}$, the collaboration may select either $0$ or $1$, but must always select the same such value; this choice
↪   has no effect on the accuracy of the classifier, so this exception can be essentially ignored. Second, the collaboration strategy need not select
↪   $\round{p_k}$ when $(p_1,p_2,\cdots,p_n)\notin (0,1)^n$---i.e., when $p_i\in \{0,1\}$ for some $i\in [n]$; this exception is somewhat more
↪   interesting, and we will return to it after stating our main result.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → PKG25NoFreeLunch.SourceCollaborationStrategy n → Prop

value:
fun {n : ℕ} (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) =>
  ∃ (k : Fin n) (alpha : PKG25NoFreeLunch.Label),
    PKG25NoFreeLunch.SourceDefersAwayFromHalf C k ∧ PKG25NoFreeLunch.SourceConstantOnHalfSlice C k alpha
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: It existentially combines one agent k and one label alpha with the displayed exact off-half and half-slice clauses. The approved nonempty-agent convention governs the source-facing uses of this definition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.SourceReliableJointLaw

Paper-local declaration:
PKG25NoFreeLunch.SourceReliableJointLaw
Source locator: cited publication, lines 37-40

Verbatim paper-source connection

```
\begin{definition}
  A collaboration strategy $\CC:[0,1]^n\rightarrow \{0,1\}$ is \textbf{reliable} if $\acc_\CC(S) \ge \min_{i\in [n]} \acc_i(S)$ for all collaboration settings
  ↪  $S$.
\end{definition}
In words, a collaboration strategy is reliable if and only if it always performs at least as well as the \textit{least accurate} agent.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → [Nonempty (Fin n)] → PKG25NoFreeLunch.SourceCollaborationStrategy n → Prop

value:
fun {n : ℕ} [Nonempty (Fin n)] (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) =>
  ∀ (S : PKG25NoFreeLunch.JointLawCollaborationSetting n),
    PKG25NoFreeLunch.SourceStrategyWellFormed S C → ∃ (i : Fin n), S.agentAccuracy i ≤ S.sourceStrategyAccuracy C
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: For every well-formed setting, existence of an agent accuracy no greater than strategy accuracy is equivalent to exceeding the minimum over the approved nonempty agent domain; the displayed joint-law and integrability conventions apply.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.SourceStrategyWellFormed

Paper-local declaration:
PKG25NoFreeLunch.SourceStrategyWellFormed
Source locator: cited publication, lines 35

Verbatim paper-source connection

```
Let $\acc_\CC(S)$ denote the 0-1 accuracy of $\Yhat_\CC$ on $\DD$. In other words, for each $x\in \XX$, each agent $i$ produces a predicted probability
↪   $P_i(x).$ The collaboration strategy aggregates these predictions into a binary classification.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → PKG25NoFreeLunch.JointLawCollaborationSetting n → PKG25NoFreeLunch.SourceCollaborationStrategy n → Prop

value:
fun {n : ℕ} (S : PKG25NoFreeLunch.JointLawCollaborationSetting n)
    (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) =>
  MeasureTheory.Integrable
    (fun (z : S.X × PKG25NoFreeLunch.Label) => if S.sourceStrategyClassifier C z.1 = z.2 then 1 else 0) S.joint
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The target is integrability of exactly the induced classifier's 0-1 correctness indicator under the joint law, which is the displayed approved expectation-definedness convention.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.UnitCubeProfile

Paper-local declaration:
PKG25NoFreeLunch.UnitCubeProfile
Source locator: cited publication, lines 27-35

Verbatim paper-source connection

```
A collaboration strategy is a way to combine the predicted probabilities of each agent to make a classification.
\begin{definition}
A \textbf{collaboration strategy} is a deterministic function $\CC: [0,1]^n \rightarrow \{0,1\}.$
\end{definition}
Here, $\CC$ should be interpreted as a function that takes in $n$ predicted probabilities and returns a 0-1 classification. Given a collaboration setting
↪    $S=(\DD, P_1, \cdots, P_n)$, a collaboration strategy $\CC$ induces the classifier
\begin{equation}
    \Yhat_\CC: \XX \rightarrow \{0,1\},\quad x \mapsto \CC(P_1(x),\cdots,P_n(x)).
\end{equation}
Let $\acc_\CC(S)$ denote the 0-1 accuracy of $\Yhat_\CC$ on $\DD$. In other words, for each $x\in \XX$, each agent $i$ produces a predicted probability
↪    $P_i(x).$ The collaboration strategy aggregates these predictions into a binary classification.
```

Lean-elaborated paper signature

```
type:
ℕ → Type

value:
fun (n : ℕ) => { p : Fin n → ℝ // ∀ (i : Fin n), 0 ≤ p i ∧ p i ≤ 1 }
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The subtype imposes exactly 0 <= p_i <= 1 for every coordinate, matching the source domain [0,1]^n used by every displayed source-facing strategy application.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.extendSourceStrategy

Paper-local declaration:
PKG25NoFreeLunch.extendSourceStrategy
Source locator: cited publication, lines 27-35

Verbatim paper-source connection

```
A collaboration strategy is a way to combine the predicted probabilities of each agent to make a classification.
\begin{definition}
A \textbf{collaboration strategy} is a deterministic function $\CC: [0,1]^n \rightarrow \{0,1\}.$
\end{definition}
Here, $\CC$ should be interpreted as a function that takes in $n$ predicted probabilities and returns a 0-1 classification. Given a collaboration setting $S=(\DD, P_1, \cdots, P_n)$, a collaboration strategy $\CC$ induces the classifier
\begin{equation}
    \Yhat_\CC: \XX \rightarrow \{0,1\},\quad x \mapsto \CC(P_1(x),\cdots,P_n(x)).
\end{equation}
Let $\acc_\CC(S)$ denote the 0-1 accuracy of $\Yhat_\CC$ on $\DD$. In other words, for each $x\in \XX$, each agent $i$ produces a predicted probability $P_i(x).$ The collaboration strategy aggregates these predictions into a binary classification.
```

Lean-elaborated paper signature

```
type:
{n : ℕ} → PKG25NoFreeLunch.SourceCollaborationStrategy n → (Fin n → ℝ) → PKG25NoFreeLunch.Label

value:
fun {n : ℕ} (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) (a : Fin n → ℝ) =>
  if hp : ∀ (i : Fin n), 0 ≤ a i ∧ a i ≤ 1 then C ⟨a, hp⟩ else false
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: On every source-domain cube profile the target is exactly C; its false branch applies only outside [0,1]^n, where the selected source definition assigns no strategy behavior.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.jointLawDependentPartitionSetting

Paper-local declaration:
PKG25NoFreeLunch.jointLawDependentPartitionSetting
Source locator: cited publication, lines 127-132

Verbatim paper-source connection

```
\paragraph{Building Calibrated Predictors from Partitions.}
Finally, given a distribution $\DD$ over $\XX\times \{0,1\}$, we show how partitions of the input space $\XX$ induce calibrated predictors. Indeed, let
↪    $\AA_i$ be a partition of $\XX$. Then $\AA_i$ induces a calibrated predictor $P_i$, where for $x\in A \in \AA_i$,
\begin{equation}
    P_i(x) := \Pr_{(X,Y)\sim \DD}[Y=1\,|\,X\in A].
\end{equation}
Here $P_i$ is the Bayes-optimal predictor given a ``coarsening'' of the input space into the partitions $\XX.$ In this way, a collaboration setting may also
↪    be identified by an ordered tuple $(\DD, \AA_1, \cdots, \AA_n).$ This approach is central to the subsequent proofs.
```

Settled source reading

The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

```
type:
{n : ℕ} →
  {Cell : Fin n → Type u_1} →
    [(i : Fin n) → Fintype (Cell i)] →
      [inst : (i : Fin n) → MeasurableSpace (Cell i)] →
        [∀ (i : Fin n), MeasurableSingletonClass (Cell i)] →
          {X : Type} →
            [inst_2 : MeasurableSpace X] →
              (joint : MeasureTheory.Measure (X × PKG25NoFreeLunch.Label)) →
                [MeasureTheory.IsProbabilityMeasure joint] →
                  (cell : (i : Fin n) → X → Cell i) →
                    (∀ (i : Fin n), Measurable (cell i)) → PKG25NoFreeLunch.JointLawCollaborationSetting n

value:
fun {n : ℕ} {Cell : Fin n → Type u_1} [(i : Fin n) → Fintype (Cell i)] [(i : Fin n) → MeasurableSpace (Cell i)]
    [∀ (i : Fin n), MeasurableSingletonClass (Cell i)] {X : Type} [MeasurableSpace X]
    (joint : MeasureTheory.Measure (X × PKG25NoFreeLunch.Label)) [MeasureTheory.IsProbabilityMeasure joint]
    (cell : (i : Fin n) → X → Cell i) (hcell : ∀ (i : Fin n), Measurable (cell i)) =>
  { X := X, measurableSpaceX := inferInstance, joint := joint,
    isProbability := PKG25NoFreeLunch.jointLawDependentPartitionSetting._proof_1 joint,
    pred := fun (i : Fin n) => PKG25NoFreeLunch.jointLawPartitionPredictor joint (cell i),
    pred_range := PKG25NoFreeLunch.jointLawDependentPartitionSetting._proof_2 joint cell,
    pred_measurable := PKG25NoFreeLunch.jointLawDependentPartitionSetting._proof_3 joint cell hcell,
    calibrated_events := PKG25NoFreeLunch.jointLawDependentPartitionSetting._proof_4 joint cell hcell }
```

Recorded source-to-declaration screening

Verdict: matches
Reason: Given a probability law and one measurable finite partition map per agent, the declaration constructs the source collaboration setting whose predictor is each cell's conditional true-label probability and proves its range, measurability, and calibration. These are the explicit finite/measurable conditions needed to make the source partition construction well-defined.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.jointLawPartitionPredictor

Paper-local declaration:
PKG25NoFreeLunch.jointLawPartitionPredictor
Source locator: cited publication, lines 127-132

Verbatim paper-source connection

```
\paragraph{Building Calibrated Predictors from Partitions.}
Finally, given a distribution $\DD$ over $\XX\times \{0,1\}$, we show how partitions of the input space $\XX$ induce calibrated predictors. Indeed, let
↪    $\AA_i$ be a partition of $\XX$. Then $\AA_i$ induces a calibrated predictor $P_i$, where for $x\in A \in \AA_i$,
\begin{equation}
    P_i(x) := \Pr_{(X,Y)\sim \DD}[Y=1\,|\,X\in A].
\end{equation}
Here $P_i$ is the Bayes-optimal predictor given a ``coarsening'' of the input space into the partitions $\XX.$ In this way, a collaboration setting may also
↪    be identified by an ordered tuple $(\DD, \AA_1, \cdots, \AA_n).$ This approach is central to the subsequent proofs.
```

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

```
type:
{X : Type u_1} →
  {Cell : Type u_2} →
    [inst : MeasurableSpace X] → MeasureTheory.Measure (X × PKG25NoFreeLunch.Label) → (X → Cell) → X → ℝ

value:
fun {X : Type u_1} {Cell : Type u_2} [MeasurableSpace X] (joint : MeasureTheory.Measure (X × PKG25NoFreeLunch.Label))
    (cell : X → Cell) (x : X) =>
  PKG25NoFreeLunch.jointLawPartitionCellPrediction joint cell (cell x)
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The value is the conditional true-label mass divided by the mass of the partition cell containing x, exactly the source's partition-induced predictor formula. Lean assigns zero on a zero-mass cell, an observationally irrelevant convention where the printed conditional probability is undefined.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.roundProb

Paper-local declaration:
PKG25NoFreeLunch.roundProb
Source locator: cited publication, lines 21-24

Verbatim paper-source connection

```
The predictor $P_i$ induces the classifier $\Yhat_i: x\mapsto \round{P_i(x)},$ where $\round{\cdot}$ is the rounding operator (without loss of generality,
↪    set $\round{0.5}=1$). $\Yhat_i$ achieves 0-1 accuracy
\begin{equation}
    \acc_i(S) := \EE_{(X,Y)\sim \DD}[\max\{P_i(X), 1 - P_i(X)\}].
\end{equation}
```

Lean-elaborated paper signature

```
type:
ℝ → PKG25NoFreeLunch.Label

value:
fun (p : ℝ) => decide (1 / 2 ≤ p)
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The declaration returns true exactly when p is at least one half, so it is the source's binary rounding operator with round(0.5)=1.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.source_definition_agree_on

Paper-local declaration:
PKG25NoFreeLunch.source_definition_agree_on
Source locator: cited publication, lines 83-92

Verbatim paper-source connection

```
\begin{definition}\label{def:correct}
For a collaboration setting $(\DD, P_1,\cdots,P_n)$ and $x\in \XX,$ we say that
\begin{itemize}
    \item agent $i$ is \textbf{correct} on $x$ if $$\Yhat_i(x) = \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agent $i$ is \textbf{incorrect} on $x$ if $$\Yhat_i(x) \neq \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agents $i$ and $j$ \textbf{agree} on $x$ if $$\Yhat_i(x) = \Yhat_j(x)$$
    \item agents $i$ and $j$ \textbf{disagree} on $x$ if $$\Yhat_i(x) \neq \Yhat_j(x).$$
\end{itemize}
For each of these statements, we can replace $i$ or $j$ with a collaboration strategy $\CC$.
\end{definition}
```

Lean-elaborated paper signature

```
type:
PKG25NoFreeLunch.Label → PKG25NoFreeLunch.Label → Prop

value:
fun (prediction1 prediction2 : PKG25NoFreeLunch.Label) => prediction1 = prediction2
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The predicate is equality of the two binary predictions, exactly the source definition of pointwise agreement for agents or a collaboration strategy.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.source_definition_correct_on

Paper-local declaration:
PKG25NoFreeLunch.source_definition_correct_on
Source locator: cited publication, lines 83-92

Verbatim paper-source connection

```
\begin{definition}\label{def:correct}
For a collaboration setting $(\DD, P_1,\cdots,P_n)$ and $x\in \XX,$ we say that
\begin{itemize}
    \item agent $i$ is \textbf{correct} on $x$ if $$\Yhat_i(x) = \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agent $i$ is \textbf{incorrect} on $x$ if $$\Yhat_i(x) \neq \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agents $i$ and $j$ \textbf{agree} on $x$ if $$\Yhat_i(x) = \Yhat_j(x)$$
    \item agents $i$ and $j$ \textbf{disagree} on $x$ if $$\Yhat_i(x) \neq \Yhat_j(x).$$
\end{itemize}
For each of these statements, we can replace $i$ or $j$ with a collaboration strategy $\CC$.
\end{definition}
```

Lean-elaborated paper signature

```
type:
PKG25NoFreeLunch.Label → ℝ → Prop

value:
fun (prediction : PKG25NoFreeLunch.Label) (eta : ℝ) => prediction = PKG25NoFreeLunch.roundProb eta
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The predicate says that the supplied binary prediction equals the rounded conditional true-label probability, exactly the source definition of pointwise correctness.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PKG25NoFreeLunch.source_definition_disagree_on

Paper-local declaration:
PKG25NoFreeLunch.source_definition_disagree_on
Source locator: cited publication, lines 83-92

Verbatim paper-source connection

```
\begin{definition}\label{def:correct}
For a collaboration setting $(\DD, P_1,\cdots,P_n)$ and $x\in \XX,$ we say that
\begin{itemize}
    \item agent $i$ is \textbf{correct} on $x$ if $$\Yhat_i(x) = \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agent $i$ is \textbf{incorrect} on $x$ if $$\Yhat_i(x) \neq \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agents $i$ and $j$ \textbf{agree} on $x$ if $$\Yhat_i(x) = \Yhat_j(x)$$
    \item agents $i$ and $j$ \textbf{disagree} on $x$ if $$\Yhat_i(x) \neq \Yhat_j(x).$$
\end{itemize}
For each of these statements, we can replace $i$ or $j$ with a collaboration strategy $\CC$.
\end{definition}
```

Lean-elaborated paper signature

```
type:
PKG25NoFreeLunch.Label → PKG25NoFreeLunch.Label → Prop

value:
fun (prediction1 prediction2 : PKG25NoFreeLunch.Label) => prediction1 ≠ prediction2
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The predicate is inequality of the two binary predictions, exactly the source definition of pointwise disagreement for agents or a collaboration strategy.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PKG25NoFreeLunch.source_definition_incorrect_on

Paper-local declaration:
PKG25NoFreeLunch.source_definition_incorrect_on
Source locator: cited publication, lines 83-92

Verbatim paper-source connection

```
\begin{definition}\label{def:correct}
For a collaboration setting $(\DD, P_1,\cdots,P_n)$ and $x\in \XX,$ we say that
\begin{itemize}
    \item agent $i$ is \textbf{correct} on $x$ if $$\Yhat_i(x) = \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agent $i$ is \textbf{incorrect} on $x$ if $$\Yhat_i(x) \neq \round{\Pr_{(X,Y)\sim \DD}[Y=1\,|\,X=x]}$$
    \item agents $i$ and $j$ \textbf{agree} on $x$ if $$\Yhat_i(x) = \Yhat_j(x)$$
    \item agents $i$ and $j$ \textbf{disagree} on $x$ if $$\Yhat_i(x) \neq \Yhat_j(x).$$
\end{itemize}
For each of these statements, we can replace $i$ or $j$ with a collaboration strategy $\CC$.
\end{definition}
```

Lean-elaborated paper signature

```
type:
PKG25NoFreeLunch.Label → ℝ → Prop

value:
fun (prediction : PKG25NoFreeLunch.Label) (eta : ℝ) => prediction ≠ PKG25NoFreeLunch.roundProb eta
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The predicate says that the supplied binary prediction differs from the rounded conditional true-label probability, exactly the source definition of pointwise incorrectness.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 56-63

Verbatim source input

```
\setcounter{theorem}{0}
\begin{theorem}\label{thm:main}
  Every reliable collaboration strategy is non-collaborative.



\end{theorem}
```

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : Nonempty (Fin n)] (C : PKG25NoFreeLunch.SourceCollaborationStrategy n),
  PKG25NoFreeLunch.SourceReliableJointLaw C →
    ∃ (k : Fin n) (alpha : PKG25NoFreeLunch.Label),
      (∀ (p : PKG25NoFreeLunch.UnitCubeProfile n),
          PKG25NoFreeLunch.Interior ↑p → ↑p k ≠ 1 / 2 → C p = PKG25NoFreeLunch.roundProb (↑p k)) ∧
        ∀ (p : PKG25NoFreeLunch.UnitCubeProfile n), PKG25NoFreeLunch.Interior ↑p → ↑p k = 1 / 2 → C p = alpha
```

Lean proof endpoint (built separately)

```
PKG25NoFreeLunch.source_theorem1_no_free_lunch_provesSpec
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: With the displayed source-domain, nonempty-agent, joint-law, and expectation conventions, the target expands non-collaboration into exactly one k and alpha with the source's off-half rounding and half-slice constancy clauses.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 100-107

Verbatim source input

```
\begin{proposition}\label{prop:linear-combination} \textbf{Linear combinations of settings.}
    Consider $\ell$ collaboration settings $S_1, \cdots, S_\ell$. Then for all $(\lambda_1, \cdots, \lambda_\ell)\in \Delta^\ell$, there exists a collaboration
    ↪    setting $S$ such that
    \begin{align}
        \acc_i(S) &= \sum_{m=1}^\ell \lambda_m \acc_i(S_m)\label{eq:monstera-1}\\
        \acc_\CC(S) &= \sum_{m=1}^\ell \lambda_m \acc_\CC(S_m)\label{eq:monstera-2}
    \end{align}
    for all $i\in [n]$ and collaboration strategies $\CC$.
\end{proposition}
```

Expanded PaperInterface specification

```
∀ {n ell : ℕ} (S : Fin ell → PKG25NoFreeLunch.JointLawCollaborationSetting n) (w : Fin ell → ℝ),
  (∀ (r : Fin ell), 0 ≤ w r) →
    ∑ r : Fin ell, w r = 1 →
      ∃ (Smix : PKG25NoFreeLunch.JointLawCollaborationSetting n),
        (∀ (i : Fin n), Smix.agentAccuracy i = ∑ r : Fin ell, w r * (S r).agentAccuracy i) ∧
          ∀ (C : PKG25NoFreeLunch.SourceCollaborationStrategy n),
            (∀ (r : Fin ell), PKG25NoFreeLunch.SourceStrategyWellFormed (S r) C) →
              PKG25NoFreeLunch.SourceStrategyWellFormed Smix C ∧
                Smix.sourceStrategyAccuracy C = ∑ r : Fin ell, w r * (S r).sourceStrategyAccuracy C
```

Lean proof endpoint (built separately)

```
PKG25NoFreeLunch.source_proposition6_linear_combination_provesSpec
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Coordinatewise nonnegative weights summing to one express the source simplex. The displayed target gives exactly the source's weighted agent and strategy accuracies; its well-formedness premise is exactly the displayed approved expectation-definedness convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 152-154

Verbatim source input

```
\begin{proposition}\label{prop:part1}
    For a collaboration strategy $\CC$, if $\acc_\CC(S)\ge \min_{i\in [n]}\acc_i(S)$ for all collaboration settings $S$, then there must exist $k\in [n]$ such
 ↪    that for all $(p_1,p_2,\cdots,p_n)\in (0,1)^n$ where $p_k\neq \frac{1}{2},$ $\CC(p_1,p_2,\cdots,p_n)=\round{p_k}.$
\end{proposition}
```

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : Nonempty (Fin n)] (C : PKG25NoFreeLunch.SourceCollaborationStrategy n),
  PKG25NoFreeLunch.SourceReliableJointLaw C →
    ∃ (k : Fin n),
      ∀ (p : PKG25NoFreeLunch.UnitCubeProfile n),
        PKG25NoFreeLunch.Interior ↑p → ↑p k ≠ 1 / 2 → C p = PKG25NoFreeLunch.roundProb (↑p k)
```

Lean proof endpoint (built separately)

```
PKG25NoFreeLunch.source_proposition7_fixed_deferral_provesSpec
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Under the displayed nonempty-agent convention, the target states reliability implies one k whose rounded prediction is followed at every interior unit-cube profile with p_k != 1/2, exactly the selected source proposition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 162-169

Verbatim source input

```
\begin{lemma}\label{lem:counterexample}
    Consider a collaboration strategy $\CC$. Suppose that there exists a tuple $(p_1,p_2,\cdots,p_n)\in (0,1)^n$ where $p_k\neq \frac{1}{2}$ and
    ↪    $\CC(p_1,p_2,\cdots,p_n)\neq \round{p_k}.$ Then there exists a collaboration setting $S_k$ such that
    \begin{itemize}
        \item[(i)] $\acc_k(S_k)>\acc_\CC(S_k)$,
        \item[(ii)]
        $\acc_i(S_k)\ge \acc_\CC(S_k)$ for all $i\neq k$.
    \end{itemize}
\end{lemma}
```

Settled source reading

The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Expanded PaperInterface specification

```
∀ {n : ℕ} {C : PKG25NoFreeLunch.SourceCollaborationStrategy n} {p : Fin n → ℝ} (hp : PKG25NoFreeLunch.Interior p)
  {k : Fin n},
  p k ≠ 1 / 2 →
    C (PKG25NoFreeLunch.Interior.toUnitCubeProfile p hp) ≠ PKG25NoFreeLunch.roundProb (p k) →
      ∃ (S : PKG25NoFreeLunch.JointLawCollaborationSetting n),
        PKG25NoFreeLunch.SourceStrategyWellFormed S C ∧
          S.sourceStrategyAccuracy C < S.agentAccuracy k ∧ ∀ (i : Fin n), S.sourceStrategyAccuracy C ≤ S.agentAccuracy i
```

Lean proof endpoint (built separately)

```
PKG25NoFreeLunch.source_lemma8_bad_tuple_provesSpec
```

Recorded source-to-Spec screening

Verdict: matches

Reason: The interior bad-profile premise, off-half restriction, and differing rounded label are exact. The strict gap for k entails the source's weak inequality at k, so the displayed weak inequality for every i is equivalent to the source's i != k clause; the displayed finite-witness, joint-law, and expectation conventions apply.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 241-246

Verbatim source input

```
\begin{proposition}\label{prop:part2}
    Consider a collaboration strategy $\CC$ such that there exists $k\in [n]$ such that for all $(p_1,p_2,\cdots,p_n)\in (0,1)^n$ where $p_k\neq
    ↪   \frac{1}{2},$ $\CC(p_1,p_2,\cdots,p_n)=\round{p_k}.$ Then, if $\acc_\CC(S)\ge \min_{i\in [n]}\acc_i(S)$ for all collaboration settings $S$, there
    ↪   must exist $\alpha\in \{0,1\}$ such that for all $(p_1,p_2,\cdots,p_n)\in (0,1)^n$ where $p_k=\frac{1}{2},$
    \begin{equation}\label{eq:horse}
        \CC(p_1,p_2,\cdots,p_n)=\alpha.
    \end{equation}
\end{proposition}
```

Settled source reading

The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Expanded PaperInterface specification

```
∀ {n : ℕ} [inst : Nonempty (Fin n)] (C : PKG25NoFreeLunch.SourceCollaborationStrategy n) (k : Fin n),
  PKG25NoFreeLunch.SourceReliableJointLaw C →
    (∀ (p : PKG25NoFreeLunch.UnitCubeProfile n),
        PKG25NoFreeLunch.Interior ↑p → ↑p k ≠ 1 / 2 → C p = PKG25NoFreeLunch.roundProb (↑p k)) →
      ∃ (alpha : PKG25NoFreeLunch.Label),
        ∀ (p : PKG25NoFreeLunch.UnitCubeProfile n), PKG25NoFreeLunch.Interior ↑p → ↑p k = 1 / 2 → C p = alpha
```

Lean proof endpoint (built separately)

```
PKG25NoFreeLunch.source_proposition9_constant_tie_label_provesSpec
```

Recorded source-to-Spec screening

Verdict: matches

Reason: Under the displayed nonempty-agent and expectation conventions, the target has the exact source premises—reliability and fixed off-half deferral to k—and concludes one alpha on every interior profile with p_k = 1/2.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation